

Dimension Reduction in Importance Sampling: Balancing Concentration and Exploration through Variable Selection

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Abstract.

This study introduces a new dimension reduction method in importance sampling for stochastic simulation, addressing the curse of dimensionality inherent in traditional approaches. Grounded in the parsimony principle, we introduce a new metric tailored to the importance sampling framework for effective variable selection. To evaluate this metric, we devise a cross-validation procedure that accounts for the unique challenges present within the importance sampling context. The proposed method strikes a critical balance between concentrating on key variables and exploring broader input areas, thereby reducing variance and enhancing the robustness of estimated outcomes. While broadly applicable, we demonstrate its effectiveness in a nonparametric importance sampling setting. Additionally, coupled with sensitivity analysis, we quantify the role and significance of each selected variable. Numerical experiments and a wind turbine case study showcase the superior performance of our approach, outperforming existing methods.

Key words: Monte Carlo sampling, reliability, variance reduction, stochastic simulation

1. Introduction

Digitization has greatly accelerated reliability-based system design, particularly in life testing. To prevent catastrophic failures over a system's lifespan, comprehensive prototype testing is essential. Advances in numerical algorithms, combined with increased computing power, now enable the development of physics-based digital computer models, paving the way for virtual testing in simulated environments.

Reliability studies in stochastic simulations predominantly employ two approaches to estimate system performance: Monte Carlo sampling (MCS) and emulator-based techniques. The emulator-

based approach constructs a surrogate model designed to approximate the response surface, allowing system performance estimations to be derived from this model (Dubourg et al. 2013). While effective for capturing the average behavior across the input space, emulator-based methods may be inadequate for estimating the probability of rare events, such as system failures, which depend on responses in localized regions (Cannamela et al. 2008). Consequently, this study adopts the MCS approach, as it provides a more robust framework for assessing the likelihood of rare events.

Among various MCS techniques, importance sampling (IS) stands out as a powerful tool, particularly due to its ability to reduce estimation variance. This variance reduction translates into enhanced computational efficiency in stochastic simulations, as larger datasets generally lead to improved accuracy. IS employs proposal (or instrumental) density to select input conditions that impose higher stress, similar to applying harsher operating conditions in physical accelerated life testing. By intelligently guiding test conditions, IS can induce more degraded system states or failure events, facilitating precise estimations of failure probability.

However, IS is prone to the curse of dimensionality as the number of factors affecting the system response grows (Cao and Choe 2019). On the other hand, it is often observed that the output of engineering systems is primarily governed by a small number of critical variables, a notion underscored by the *parsimony principle*. Leveraging this principle, this study introduces a new dimension reduction approach for IS within the context of reliability evaluation.

One important insight we discover in this study is the need to balance concentration and exploration. Given that the underlying premise of IS is to direct simulation efforts toward the most influential input conditions, a seemingly intuitive approach to dimension reduction is to identify crucial input variables and design a proposal density centered around them. However, our findings suggest that an excessive focus on these variables may backfire, potentially increasing estimation variance. The trade-off between concentration and exploration is akin to the exploitation-exploration balance often considered in decision-making problems, such as multi-armed bandit scenarios. In the IS context, concentrating on key variables corresponds to exploiting narrow input conditions that frequently lead to the event of interest, whereas considering additional input variables involves exploring broader input regions.

In this study, we devise a new method to obtain a balanced proposal density with a carefully selected subset of input variables, referred to as *IS with Variable Selection* (IS-VS). This approach offers a comprehensive overview of how the selected variables influence the final results. Our contributions are summarized as follows.

- To the best of our knowledge, this study is the first to explore dimension reduction within the IS framework. We derive an optimal proposal density constructed using only a subset of variables.
- Our investigation highlights the critical need to balance concentration with sufficient exploration. While existing variable selection criteria, such as AIC and BIC, are commonly used in machine learning to achieve a balanced set of input variables, our goal of minimizing variance in estimating tail probability differs from typical subset selection problems. As a result, we develop a new metric for variable selection, specifically tailored to the IS framework.
- To evaluate the metric, we devise a new cross-validation (CV) procedure. While our approach preserves the core principles of CV, it addresses the unique computational challenges specific to the IS context.

We also conduct a sensitivity analysis to quantify the importance of each variable and clarify their roles in balancing concentration and exploration. Interestingly, our findings reveal that the selected variables fall into two categories: (1) key essential variables that substantially affect the system response, enabling concentrated simulation efforts; (2) seemingly less-important supplementary variables that facilitate exploration. The superior performance of IS-VS in numerical examples and wind turbine case study demonstrate that by striking the right balance, our approach significantly enhances the capability for variance reduction, ultimately improving the robustness of the estimated outcomes.

The paper is organized as follows. Section 2 introduces the background of IS. Section 3 reviews studies on IS and dimension reduction. Section 4 discusses the proposed IS-VS approach. Section 5 presents the application of the proposed approach in the nonparametric setting. Section 6 implements IS-VS using several numerical examples and compares its performance in comparison with other alternatives. Section 7 presents the wind turbine case study. Lastly, Section 8 concludes the paper. Additionally, the online supplementary document includes derivations and proof, summary of notations, and additional experimental results.

2. Background

Consider a digital computer model with the output $Y \in \mathbb{R}^1$ at the input $\mathbf{X} \in \mathbb{D} \subseteq \mathbb{R}^d$, where $\mathbb{D}$ is the domain of $\mathbf{X}$, and $\mathbf{X}$ follows a known distribution $f(\mathbf{x})$. We are interested in estimating the expectation of function of the simulation outcome Y , that is, $E[g(Y)]$ for some function $g(\cdot)$. In particular, with $g(Y) = \mathbb{I}(Y > l)$ where $\mathbb{I}(\cdot)$ denotes an indicator function and l represents a resistance level, $E[g(Y)]$ becomes the failure probability $P(Y > l)$, also referred to as the probability of

exceedance (POE) (Choe et al. 2016). While IS can be generally employed to any form of $g(\cdot)$, we focus our discussion on the POE estimation in subsequent discussions.

The Crude Monte Carlo (CMC) method samples the input $\mathbf{X}$ from f and runs the computer model at the sampled $\mathbf{X}$ to get the simulation output Y . While simple and easy to implement, CMC is known to result in large variance in estimation. Variance reduction techniques have been employed to reduce the computational efforts required to achieve a certain level of variance or uncertainty. Some common techniques include control variate (Nelson 1990), which adjusts the estimate by accounting for the covariance with an additional auxiliary variable, and stratified sampling (Singh and Mangat 1996, Liu et al. 2022, Jain et al. 2025), which partitions the input space into a finite number of strata and draws different numbers of samples from each stratum.

Among the variance reduction techniques, IS proves effective in many applications (Jia and Gardoni 2018). Unlike CMC, which draws samples from the original density f , IS operates by drawing samples from a proposal distribution q , instead of f , and estimates the POE by

$$\hat{P}_{IS} = \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j) \frac{f(\mathbf{X}_j)}{q(\mathbf{X}_j)}, \quad (1)$$

where $\mathbf{X}_j$ and Y_j are the j -th sampled input and simulated outcome, respectively, and N_T is the total number of samples. Note that the likelihood ratio f/q is employed to ensure unbiasedness. The proposal density q is also referred to as an instrumental density. $\hat{P}_{IS}$ is unbiased if $q(\mathbf{x}) = 0$ implies that $\mathbb{I}(Y > l | \mathbf{X} = \mathbf{x}) f(\mathbf{x}) = 0$ for all $\mathbf{x}$, implying that q must cover all regions where failure events can possibly occur (Choe et al. 2015).

IS aims to minimize the variance of $\hat{P}_{IS}$ by drawing samples from the most important regions. The optimal IS density depends on the type of computer models. For deterministic (or noiseless) computer models that simulate a fixed output Y at a given input, the optimal density is given by

$$q_{DIS} = \frac{1}{C_{DIS}} \mathbb{I}(Y > l | \mathbf{X} = \mathbf{x}) f(\mathbf{x}), \quad (2)$$

where $C_{DIS} = \int_{\mathbb{D}} \mathbb{I}(Y > l | \mathbf{X} = \mathbf{x}) f(\mathbf{x}) d\mathbf{x}$ is the normalizing constant (Kurtz and Song 2013). The subscript *DIS* stands for the IS for the deterministic computer model. Despite the deterministic nature of these computer models, such simulations are still termed stochastic simulations in the literature, because inputs are stochastic (Glynn and Iglehart 1989). With q_{DIS} , the estimator $\hat{P}_{IS}$ in (1) remains unbiased for $P(Y > l | \mathbf{X} = \mathbf{x})$, since the proposal density satisfies the required support condition.

Recently, stochastic computer models, which generate noisy outputs at a given input, gain much attention (Ankenman et al. 2010). Simulations with stochastic computer models are also termed two-level or nested simulation (Choe et al. 2018). Let ϵ denote the random vector embedded (or hidden) in the black box computer model. This embedded randomness represents stochastic variability inherent in the simulation outputs. A simulation output Y is governed by both the input vector $\mathbf{X}$ and the random element ϵ . Let $f(\mathbf{X}, \epsilon)$ and $q(\mathbf{X}, \epsilon)$, respectively, denote the joint original density and proposal density of $\mathbf{X}$ and ϵ . The POE estimator becomes

$$\begin{aligned}\hat{P}_{SIS} &= \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j, \epsilon = \epsilon_j) \frac{f(\mathbf{X}_j, \epsilon_j)}{q(\mathbf{X}_j, \epsilon_j)} \\ &= \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j, \epsilon = \epsilon_j) \frac{f(\mathbf{X}_j) f(\epsilon_j | \mathbf{X}_j)}{q(\mathbf{X}_j) q(\epsilon_j | \mathbf{X}_j)}.\end{aligned}\quad (3)$$

The subscript *SIS* stands for the IS for the stochastic computer model, or shortly, stochastic IS (Choe et al. 2015, Ko and Byon 2022).

In the nested simulation model, ϵ_j is drawn internally by the simulator from its conditional distribution $f(\epsilon | \mathbf{X})$, given the sampled input $\mathbf{X} = \mathbf{X}_j$. Even in the IS setting, ϵ_j can be drawn from its original distribution $f(\epsilon | \mathbf{X})$ only due to the black-box nature of the simulator, while $\mathbf{X}_j$ can be sampled from its proposal distribution $q(\mathbf{X})$. Consequently, the only feasible choice for the proposal density $q(\epsilon | \mathbf{X})$ is $f(\epsilon | \mathbf{X})$. This highlights that importance weighting applies only to the input $\mathbf{X}$, not to the embedded randomness ϵ . Accordingly, $\hat{P}_{SIS}$ in (3) can be written as

$$\begin{aligned}\hat{P}_{SIS} &= \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j, \epsilon = \epsilon_j) \frac{f(\mathbf{X}_j) f(\epsilon_j | \mathbf{X}_j)}{q(\mathbf{X}_j) f(\epsilon_j | \mathbf{X}_j)} \\ &= \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j, \epsilon = \epsilon_j) \frac{f(\mathbf{X}_j)}{q(\mathbf{X}_j)},\end{aligned}\quad (4)$$

where $\mathbf{X}_j$ is drawn from q and ϵ_j is sampled within the computer model, given the simulation input $\mathbf{X} = \mathbf{X}_j$. This result shows that the proposal adjustment operates solely on $\mathbf{X}$.

Choe et al. (2015) derived the optimal proposal density that minimizes the variance of $\hat{P}_{SIS}$ as

$$q_{SIS}(\mathbf{x}) = \frac{1}{C_{SIS}} f(\mathbf{x}) \sqrt{s(\mathbf{x})}, \quad (5)$$

where $C_{SIS} = \int_{\mathbb{D}} f(\mathbf{x}) \sqrt{s(\mathbf{x})} d\mathbf{x}$ is the normalizing constant and $s(\mathbf{x})$ is the conditional failure probability or conditional POE at $\mathbf{X} = \mathbf{x}$ as $s(\mathbf{x}) = P(Y > l | \mathbf{X} = \mathbf{x})$. With $q_{SIS}(\mathbf{x})$, $\hat{P}_{SIS}$ in (4) is unbiased, because the support of q includes the support of $P(Y > l | \mathbf{X} = \mathbf{x})f(\mathbf{x})$ (Pan et al. 2020).

While the proposal densities in (2) and (5), respectively, represent theoretically optimal instrumental densities for deterministic and stochastic computer models, they cannot be straightforwardly implemented. This issue arises because $\mathbb{I}(\cdot)$ in (2) and $s(\cdot)$ in (5) are unknown due to the black-box nature of the computer model. Consequently, there have been a considerable attention to approximate them using parametric and nonparametric methods.

3. Literature Review

3.1. Importance Sampling

The underlying premise of IS is to reweight sampling efforts so that regions of interest receive greater emphasis through a well-designed proposal density. However, as the input dimension increases, these regions occupy an increasingly sparse fraction of the total volume. Without dimensionality reduction, the search space expands rapidly, making it particularly challenging to identify effective regions in the original input space.

Parametric IS approaches formulate the proposal distribution that depends on some parameters, aiming for the approximate proposal density to closely mimic the optimal proposal density. In general, the approximation quality depends heavily on the selected distribution type (De Boer et al. 2005). To flexibly identify the effective regions, Kurtz and Song (2013) and Cao and Choe (2019) employ a Gaussian mixture model (GMM) and obtain the parameters to minimize the Kullback–Leibler divergence criterion. Recently, Zhang et al. (2022) employed a von Mises-Fisher mixture model to handle complex response surfaces. Such mixture models are useful primarily for low-dimensional input cases. However, their performance deteriorates as the input dimension grows, since the number of component densities can rapidly escalate. Likewise, kernel-based nonparametric methods (Neddermeyer 2009, Li et al. 2021), while useful when the target distribution has an irregular shape, face similar difficulties. The number of kernel terms can increase drastically, and redundancy among kernels may cause computational inefficiency and model overfitting.

For these reasons, to our knowledge, existing IS approaches are generally limited to input dimensions of at most three to five (Cao and Choe 2019, Li et al. 2021), and previous studies have demonstrated significant performance degradation even within this range (Au and Beck 2003, Cao and Choe 2019).

One might argue that a straightforward remedy is to select a small subset of key variables. However, relying solely on these variables can cause the problem of weight degeneracy, where only a small fraction of samples carry disproportionately large weights, thus inflating the estimator variance (El-Laham et al. 2018, Vehtari et al. 2024); see Section 4 for a detailed discussion. This underscores the need for new approaches capable of handling high-dimensional inputs effectively by striking a balance between concentration and degeneracy. In particular, overcoming these limitations calls for the development of a rigorous criterion for variable selection or dimension reduction, which remains largely unexplored in the IS literature.

3.2. Dimension reduction

Typical dimension reduction and variable selection methods, such as Lasso regression or approaches based on information criteria like AIC and BIC, are primarily designed to improve prediction accuracy in unseen datasets by reducing model complexity. These approaches generally identify variables (or reduced dimensions) that contribute most to the performance of a regression or classification model in approximating a response surface such as $E(Y|\mathbf{X})$ or $P(Y > l|\mathbf{X})$, i.e., conditional response given $\mathbf{X}$. However, the goal of IS for rare event analysis is fundamentally different: the objective is to minimize the variance of the POE estimator, rather than minimizing the prediction error.

This distinction gives rise to several reasons why conventional dimension reduction methods may be ineffective in the IS context. First, existing approaches are generally designed to estimate or predict the response surface across the entire input space. In contrast, IS is primarily concerned with rare event probability estimation, where only a narrow effective region is critical, and the behavior of the distribution tails carries far greater importance than overall average-fit performance. Next, existing variable selection methods do not account for the risk of weight degeneracy, which is a unique concern in IS. By emphasizing only the most critical variables, existing methods may produce overly narrow proposal distributions, thereby increasing the likelihood of weight degeneracy.

Finally, we highlight the challenges of applying common projection-based dimension reduction methods, such as principal component analysis (PCA). Implementing PCA within IS requires knowledge of the exact distribution in the reduced space to enable valid sampling, which is rarely met in practice. In addition, mapping samples from the reduced space back to the original input space, which is necessary for performing simulations, is nontrivial and not necessarily one-to-one, potentially introducing the risk of bias.

In summary, the performance of IS is governed by the distributional impact of the selected variables on the IS estimator. As a result, variables identified by existing dimension reduction methods may not provide meaningful guidance for designing proposal densities that achieve effective variance reduction. These limitations highlight the need for a new dimension reduction criterion explicitly tailored to the objective of IS.

4. Methodology

This section presents a new methodology for dimension reduction in IS using selected variables.

4.1. Importance sampling with a subset of variables

The rationale behind using IS with selected variables is that the parsimony principle often holds in engineering systems, where a system's functioning is typically governed by a handful of crucial factors. By identifying these influential variables and constructing the proposal density with them, we aim to enhance the simulation efficiency. Suppose we divide the input vector $\mathbf{X}$ into two groups: $\mathbf{X}_M \in \mathbb{D}_M \subset \mathbb{R}^{d_M}$ denoting a vector of d_M -dimensional informative major variables, and $\mathbf{X}_{-M} \in \mathbb{D}_{-M} \subset \mathbb{R}^{d-d_M}$ a vector of $(d - d_M)$ -dimensional remaining variables. We will discuss later how to identify $\mathbf{X}_M$ from the original d -dimensional input vector $\mathbf{X}$ in Section 4.3. This section focuses on deriving the optimal proposal density with a subset of variable.

Let $\mathbf{X}_{Mj}$ and $\mathbf{X}_{-Mj}$ denote the j -th random sample of major and minor variables, respectively. The POE estimator is

$$\hat{P}_{IS-VS} = \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j) \frac{f(\mathbf{X}_j)}{q(\mathbf{X}_j)}, \quad (6)$$

where Y_j is the simulation output obtained at $\mathbf{X}_j = (\mathbf{X}_{Mj}, \mathbf{X}_{-Mj})$. When the computer model is deterministic, Y_j is uniquely determined at the input $\mathbf{X}_j$; Otherwise, it is stochastic.

Suppose we apply IS to the major variables only. In this case, we use the proposal density q_M to draw samples of the major variables $\mathbf{X}_M$ while the remaining minor variables $\mathbf{X}_{-M}$ are sampled from their original conditional density $f_{-M/M}$ (i.e., $q_{-M/M} = f_{-M/M}$). Applying the procedure similar to (3)-(4), $\hat{P}_{IS-VS}$ becomes

$$\hat{P}_{IS-VS} = \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}) \frac{f_M(\mathbf{X}_{Mj})}{q_M(\mathbf{X}_{Mj})}, \quad (7)$$

where $\mathbf{X}_{Mj} \sim q_M$ and $\mathbf{X}_{-Mj} \sim f_{-M/M}(\cdot | \mathbf{X}_{Mj})$. Note that $\mathbf{X}_{-Mj}$ continues to affect the response Y , because the simulation output Y_j is determined by both $\mathbf{X}_{Mj}$ and $\mathbf{X}_{-Mj}$. For stochastic computer models, the indicator function could be more explicitly written as $\mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}, \epsilon = \epsilon_j)$, where ϵ_j is drawn internally by the simulator from its original distribution $f(\epsilon | \mathbf{X}_j)$. The detailed discussion to derive (7) is provided in the online supplementary materials.

The result in (7) implies that $\mathbf{X}_{-M}$ can be treated as the noise term ϵ in (3) for the stochastic computer model. Since IS targets only $\mathbf{X}_M$ while the minor variables are drawn from $f_{-M/M}$, the key quantity driving variance reduction is the conditional POE at $\mathbf{X}_M = \mathbf{x}_M$, defined as

$$s_M(\mathbf{x}_M) = P(Y > l | \mathbf{X}_M = \mathbf{x}_M). \quad (8)$$

Theorem 1 obtains the optimal proposal density q_{IS-VS}^* that minimizes the variance of $\hat{P}_{IS-VS}$ in (7). It shows that, when IS is applied only to $\mathbf{X}_M$, the proposal density should be guided by the conditional failure probability $s_M(\mathbf{x}_M)$. Consequently, variance reduction is governed by the likelihood of failure at a given configuration of these variables. The proof is provided in the online supplementary materials.

THEOREM 1. *Suppose that the d -dimensional input vector $\mathbf{X}$ is divided into two groups, $\mathbf{X}_M \in \mathbb{D}_M \subset \mathbb{R}^{d_M}$ and $\mathbf{X}_{-M} \in \mathbb{D}_{-M} \subset \mathbb{R}^{d-d_M}$ for $d_M < d$. Consider the failure probability estimator $\hat{P}_{IS-VS}$ in (7), where X_{Mj} and X_{-Mj} are sampled from q_M and $f_{-M/M}$, respectively. For both deterministic and stochastic computer models, the optimal proposal density to minimize the variance of $\hat{P}_{IS-VS}$ in (7) is given by*

$$q_{IS-VS}^*(\mathbf{x}) = \frac{1}{C_q^*} f(\mathbf{x}) \sqrt{s_M(\mathbf{x}_M)}, \quad (9)$$

where C_q^* is the normalizing constant.

$\hat{P}_{IS-VS}$ in (7) is an unbiased estimator for $P(Y > l)$ under the optimal proposal density $q_{IS-VS}^*(\mathbf{x})$ in (9) (Choe et al. 2016). In practice, the conditional POE $s_M(\mathbf{x}_M)$ should be approximated. When the support of the proposal density with the approximated conditional POE covers the support of $P(Y > l | \mathbf{X}_M = \mathbf{x}_M, \mathbf{X}_{-M} = \mathbf{x}_{-M}) f_M(\mathbf{x})$, $\hat{P}_{IS-VS}$ remains unbiased (Pan et al. 2020). For instance, with the nonparametric construction presented in Section 5, the proposal density is positive for all $\mathbf{x}$, so the unbiasedness condition is satisfied.

Now, the question boils down to how to identify the informative variables $\mathbf{X}_M$. In the next section, we present a critical issue, called *weight degeneracy*, and present the method for finding an appropriate vector of major variables in Section 4.3.

4.2. Trade-off between concentration and exploration

The general notion in IS is to strategically design a proposal density to draw samples from the most relevant regions. This concept can easily lead to the temptation to opt for a more concentrated distribution under the belief that “the more focus, the better.” Consequently, this can lead to selecting a smaller number of the most important variables when constructing $\mathbf{X}_M$. However, this narrow focus can overlook the need for exploration, potentially increasing variance.

To further investigate, let us consider the cross-entropy (CE) function. Let $\hat{s}_M(\cdot)$ denote an conditional POE estimator for approximating $s_M(\cdot)$ in (8) with $\mathbf{X}_M$. The CE function evaluates the closeness of $\hat{s}_M(\cdot)$ to the observation $Z = \mathbb{I}(Y > l)$, as follows.

$$CE = - \sum_{j=1}^n Z_j \ln(\hat{s}_M(\mathbf{x}_{Mj})) + (1 - Z_j) (\ln(1 - \hat{s}_M(\mathbf{x}_{Mj}))), \quad (10)$$

where n denotes the number of data samples used for computing CE. In our implementation of the proposed IS-VS method, we adopt an iterative procedure and compute the CE at each iteration using all data samples collected up to that point.

Note that CE diminishes as $\hat{s}_M(\cdot)$ approaches Z_j more closely (Li et al. 2021). As such, a natural choice is to find $\mathbf{X}_M$ that can lead to the smallest CE. This can be done by enumeration or by efficient search strategies, such as greedy search or simulated annealing. However, selecting subsets solely based on the CE criterion emphasizes concentration of sampling efforts, and excessive concentration can backfire. In particular, with a small number of key important variables, $\hat{s}_M(\cdot)$ tends to concentrate around 0 or 1, i.e., $\hat{s}_M(\cdot)$ is closer to 0 (or 1), where Z_j is 0 (or 1), leading to a small CE . While a small CE could suggest a good estimation of conditional POE, an estimator with a small CE might concentrate too narrowly when drawing inputs. To illustrate this, let us consider Example 1 presented in Section 6, which we will discuss in detail later. In this example, X_1 and X_2 are the most important variables, while other variables are less significant. Accordingly, the estimator with X_1 and X_2 tends to achieve the lowest CE value, however, relying solely on these two variables results in large variance.

Although this may seem counterintuitive, such backlash arises due to over-concentration. A narrow focus may unduly restrict the exploration of the input space, potentially missing broader pattern that could affect the estimation accuracy. Mathematically, this phenomenon can be explained

by the issue of *weight degeneracy* (El-Laham et al. 2018). Let δ_j denote the IS weight of the j th sample, defined as $\delta_j = f_M(\mathbf{X}_{Mj})/q_M(\mathbf{X}_{Mj})$ (Vehtari et al. 2024). The POE in (7) becomes

$$\hat{P}_{IS-VS} = \frac{1}{N_T} \sum_{j=1}^{N_T} \mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}) \delta_j, \quad (11)$$

where N_T denotes the number of data samples to obtain $\hat{P}_{IS-VS}$. In the iterative implementation of the proposed IS-VS method, N_T represents the number of data samples available at each iteration. Here, for stochastic computer models, the indicator function could be more explicitly written as $\mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}, \epsilon = \epsilon_j)$.

The POE in (11) implies that the POE estimator is the weighted sum of $\mathbb{I}(\cdot)$'s. Samples with large IS weights significantly influence the estimated POE. Large weights occur when samples are drawn from regions where f_M is much larger than q_M , i.e., $f_M \gg q_M$. This happens when q_M is sharp and overly concentrated, causing a few samples to receive disproportionately large weights compared to the majority of other weights. This imbalance makes the estimator rely too heavily on a small number of samples with substantial weights, leading to high variance in the estimation. As a result, the estimates become highly sensitive to the specific samples drawn, reducing robustness and potentially causing the results to deviate significantly from true values.

4.3. Variable selection with new complexity measure

Our goal is to identify an optimal set of major variables that minimize the variance of $\hat{P}_{IS-VS}$ in (7) by effectively balancing between concentration and exploration. While a simpler model with fewer variables is generally preferred, we must avoid an over-simplification, as it may cause large variance due to weight degeneracy. Recognizing that model complexity can be measured by the number of variables, we propose the following complexity measure.

$$C_\alpha(\mathbf{X}_M) = SE(\hat{P}_M) + \alpha \mathcal{P}(d_M), \quad (12)$$

where $\hat{P}_M$ is the POE estimator with the major variables $\mathbf{X}_M$ with dimension d_M , and $SE(\hat{P}_M)$ is its standard error (SE). In the second term, $\mathcal{P}(d_M)$ serves as a penalty term that discourages excessive model complexity, while α is a positive constant that balances the trade-off between variance and complexity.

The optimal $\mathbf{X}_M$ that minimizes $C_\alpha(\mathbf{X}_M)$ depends on the value of α . Our primary analytical tool for determining α is the CV, similar to the cost complexity parameters employed in Classification and Regression Tree (Hastie et al. 2005, Liu et al. 2022). However, we cannot blindly employ the

typical CV due to the unique challenges present in IS. Specifically, special considerations must be addressed, as data must be generated from the computer model, unlike typical machine learning problems that rely on available datasets for the CV procedure.

We outline the overall procedure for evaluating $C_\alpha(\mathbf{X}_M)$ in (12). Suppose we have an available dataset simulated from the computer model, which could be a pilot dataset or one obtained from previous iterations in the iterative process. We partition this dataset and construct multiple training and validation sets through methods such as k -fold CV or by randomly dividing the dataset into two exclusive training and validation subsets several times. The procedure consists of two phases. In the first phase, we obtain the POE estimate for each estimator $\hat{s}_M(\cdot)$ using each *training set* and compute its SE (the first term in (12)) from multiple training sets. In the second phase, we choose the estimator that minimizes the complexity measure in (12) for some α . Then we gauge the performance of each chosen estimator by obtaining the SE of its POE anchored on *validation sets*. Finally, we choose the estimator with the smallest SE in the validation sets. This two-phase procedure ensures that the selected estimator minimizes out-of-sample variance. Next, we delve into each phase in greater detail, focusing on how we address the unique challenges specific to the IS context.

4.3.1. Phase 1: Evaluating the standard error with training set. Let us first consider a single training set and let $\hat{s}_M^{tr}(\cdot)$ denote the conditional POE estimator with the major variables $\mathbf{X}_M$, constructed with the training set. We design the proposal density for $\mathbf{X}_M$ as

$$q_M(\mathbf{x}_M) = \frac{1}{C_M} f_M(\mathbf{x}_M) \sqrt{\hat{s}_M^{tr}(\mathbf{x}_M)}, \quad (13)$$

where $f_M(\mathbf{x}_M)$ denotes the density of $\mathbf{X}_M$. The POE estimate in the training set becomes

$$\hat{P}_{tr} = \frac{1}{n_{tr}} \sum_{j=1}^{n_{tr}} \mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}) \frac{f_M(\mathbf{X}_{Mj})}{q_M(\mathbf{X}_{Mj})}. \quad (14)$$

Here, $\mathbf{X}_{Mj}$ is the j -th sample drawn from q_M in (13), Y_j is the j -th simulation output for $j = 1, 2, \dots, n_{tr}$, and n_{tr} is the number of data points in the training set. For stochastic computer models, the indicator function could be more explicitly written as $\mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}, \epsilon = \epsilon_j)$.

The challenge, however, is that $\hat{P}_{tr}$ in (14) is not readily available. This presents a key difference in the CV procedure between the IS and typical machine learning contexts. In order to evaluate the performance of q_M , $\mathbf{X}_{Mj}$ and $\mathbf{X}_{-Mj}$ should be sampled from q_M and $f_{-M/M}$, respectively, and the corresponding Y_j in (14) should be obtained. However, it is possible that such data points are

not available in the training set. Instead, one would need to run the computer model n_{tr} times, which adds substantial computational overhead. Conversely, without running the computer model, Y_j cannot be obtained.

To address the problem while utilizing the information in the training set, we approximate the simulation outcome $\mathbb{I}(\cdot)$ in (14) using the estimated conditional POE $\hat{s}_M^{tr}$ obtained with the training data and approximate $\hat{P}_{tr}$ as follows.

$$\tilde{P}_{tr} = \frac{1}{n_{tr}} \sum_{j=1}^{n_{tr}} \hat{s}_M^{tr}(\mathbf{X}_{Mj}) \frac{f_M(\mathbf{X}_{Mj})}{q_M(\mathbf{X}_{Mj})}. \quad (15)$$

It has been known that the normalizing constant C_M in (13) could be unstable for multivariate density (Li et al. (2021)). Thus, we instead use the self-normalizing constant (Owen (2013)) that replaces n_{tr} with $\sum_{j=1}^{n_{tr}} f_M(\mathbf{X}_{Mj})/q_M(\mathbf{X}_{Mj})$ and consider the following self-normalized estimator.

$$\tilde{P}_{tr} = \sum_{j=1}^{n_{tr}} \frac{\hat{s}_M^{tr}(\mathbf{X}_{Mj}) f_M(\mathbf{X}_{Mj}) / q_M(\mathbf{X}_{Mj})}{\sum_{j=1}^{n_{tr}} f_M(\mathbf{X}_{Mj}) / q_M(\mathbf{X}_{Mj})}. \quad (16)$$

This self-normalizing estimator is known to be asymptotically unbiased, and for finite sample sizes, the variance reduction it achieves often outweighs the small bias it introduces (Owen 2013, Hesterberg 1988, Li et al. 2021). Plugging (13) into (16) yields

$$\tilde{P}_{tr} = \frac{\sum_{j=1}^{n_{tr}} \frac{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}{\sqrt{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}}}{\sum_{j=1}^{n_{tr}} \frac{1}{\sqrt{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}}} = \frac{\sum_{j=1}^{n_{tr}} \sqrt{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}}{\sum_{j=1}^{n_{tr}} \frac{1}{\sqrt{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}}}. \quad (17)$$

We repeat the procedure with multiple training sets. With each training set, we obtain $\tilde{P}_{tr}$ in (17). We then compute the SE of these $\tilde{P}_{tr}$ values, setting it as $SE(\hat{P}_M)$ in $C_\alpha(\mathbf{X}_M)$ in (12).

4.3.2. Phase 2: Choosing the best subset of input variables with validation set. The penalty parameter α in (12) plays an important role. With large α , the complexity measure in $C_\alpha(\mathbf{X}_M)$ aggressively penalizes the number of variables and chooses $\mathbf{X}_M$ with a small number of variables, leading to the risk of weight degeneracy. On the other hand, $C_\alpha(\mathbf{X}_M)$ with small α would emphasize the first term over the second term, potentially resulting in an estimator with unnecessarily many variables, i.e., an overly complicated proposal density that fails to properly concentrate simulation efforts.

A reasonable remedy is to try multiple values for α and choose the estimator that provides the lowest SE in the validation set. However, determining the range of α and listing its candidate values

is not immediately clear. It depends on the order of the first term $SE(\hat{P}_M)$ in $C_\alpha(\mathbf{X}_M)$, which is unknown beforehand. Lemma 1 shows the conditions under which an estimator $\hat{s}_M(\cdot)$ with $\mathbf{X}_M$ can minimize the complexity measure for some α .

LEMMA 1. Suppose that $SE(\hat{P}_M)$ with $\mathbf{X}_M$ of dimension d_M satisfies

$$\max_{\{\mathbf{X}'_M: \mathcal{P}(d_M) < \mathcal{P}(d'_M)\}} \frac{SE(\hat{P}_M) - SE(\hat{P}'_M)}{\mathcal{P}(d'_M) - \mathcal{P}(d_M)} \leq \min_{\{\mathbf{X}'_M: \mathcal{P}(d_M) > \mathcal{P}(d'_M)\}} \frac{SE(\hat{P}'_M) - SE(\hat{P}_M)}{\mathcal{P}(d_M) - \mathcal{P}(d'_M)}, \quad (18)$$

and

$$\min_{\{\mathbf{X}'_M: \mathcal{P}(d_M) > \mathcal{P}(d'_M)\}} \frac{SE(\hat{P}'_M) - SE(\hat{P}_M)}{\mathcal{P}(d_M) - \mathcal{P}(d'_M)} \geq 0, \quad (19)$$

where $\hat{P}'_M$ denotes the POE estimator with the major variable $\mathbf{X}'_M$ with dimension d'_M . Further, suppose that it holds

$$SE(\hat{P}_M) \leq SE(\hat{P}'_M) \quad (20)$$

for $\mathcal{P}(d'_M) = \mathcal{P}(d_M)$, $\forall \mathbf{X}'_M \neq \mathbf{X}_M$. Then there exists α ($\alpha \geq 0$) such that $C_\alpha(\mathbf{X}_M) \leq C_\alpha(\mathbf{X}'_M)$, $\forall \mathbf{X}'_M \neq \mathbf{X}_M$.

The result in Lemma 1 implies that if $\mathbf{X}_M$ satisfies the conditions in Lemma 1, it minimizes the complexity measure for some α . Consequently, even without explicitly determining such α value, we can consider $\mathbf{X}_M$ as the candidate for the next CV phase. This means that we do not need to identify the range of α or test multiple α values for CV. Instead, we only need to verify whether each $\mathbf{X}_M$ satisfies the conditions in (18)-(20) in Lemma 1.

With the selected list of input vectors, we evaluate their performance using the validation sets. Let us first use a single validation set. Consider the proposal density of $\mathbf{X}_M$, denoted by q_M , which is constructed using the training set. Then its POE for the validation set is

$$\hat{P}_{vld} = \frac{1}{n_{vld}} \sum_{j=1}^{n_{vld}} \mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}) \frac{f_M(\mathbf{X}_{Mj})}{q_M(\mathbf{X}_{Mj})}, \quad (21)$$

where n_{vld} is the number of data points in the validation set, $\mathbf{X}_{Mj}$ is the j -th sample drawn from q_M , and Y_j is the associated simulation output for $j = 1, \dots, n_{vld}$. For stochastic computer models, the indicator function could be more explicitly written as $\mathbb{I}(Y_j > l | \mathbf{X}_M = \mathbf{X}_{Mj}, \mathbf{X}_{-M} = \mathbf{X}_{-Mj}, \epsilon = \epsilon_j)$.

Similar to the challenge encountered in Phase 1, the validation set may not include the sampled $\mathbf{X}_{Mj}$ and $\mathbf{X}_{-Mj}$ and thus, the corresponding Y_j . To utilize the information in the validation set while avoiding additional simulation runs to get Y_j 's, we approximate $\mathbb{I}(\cdot)$ in (21) with the estimated conditional POE using the data in the validation set. This is similar to what is conducted in

Phase 1, however, Phase 2 uses the validation set to approximate $\mathbb{I}(\cdot)$. One might argue that we can simply use $\hat{s}_M^{tr}(\cdot)$, the conditional POE estimate learned with the training set, to approximate $\mathbb{I}(\cdot)$ in (21). However, such attempt would incur the overfitting issue and violate the fundamental principle of CV—training a model on the training set and testing the resulting model on a separate validation set. In our case, q_M in (21) is the trained model, and we need to evaluate its out-of-sample performance using an unseen dataset. Thus, instead of recycling $\hat{s}_M^{tr}(\cdot)$, our approach newly estimates the conditional POE using the validation set as a proxy to replace $\mathbb{I}(\cdot)$ in (21).

Specifically, let $\hat{s}_M^{vld}(\cdot)$ denote the estimated conditional POE, using the validation set. We approximate $\hat{P}_{vld}$ in (21) as

$$\tilde{P}_{vld} = \frac{1}{n_{vld}} \sum_{j=1}^{n_{vld}} \hat{s}_M^{vld}(\mathbf{X}_{Mj}) \frac{f_M(\mathbf{X}_{Mj})}{q_M(\mathbf{X}_{Mj})}. \quad (22)$$

By using the self-normalizing constant that replaces n_{vld} with $\sum_{j=1}^{n_{vld}} f_M(\mathbf{X}_{Mj})/q_M(\mathbf{X}_{Mj})$, we can also consider the following self-normalized estimator.

$$\tilde{P}_{vld} = \frac{\sum_{j=1}^{n_{vld}} \frac{\hat{s}_M^{vld}(\mathbf{X}_{Mj})}{\sqrt{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}}}{\sum_{j=1}^{n_{vld}} \frac{1}{\sqrt{\hat{s}_M^{tr}(\mathbf{X}_{Mj})}}}. \quad (23)$$

We repeat the proposed procedure with multiple validation sets and choose $\mathbf{X}_M$ that has the smallest SE of $\tilde{P}_{vld}$. The variable selection algorithm is summarized in Algorithm 1. In Lines #8 and #20, we can use the acceptance-rejection sampling method to draw samples from the proposal density (Choe et al. 2015). Let $\mathbf{X}_M^*$ denote the finally chosen major variables with its associated estimator $\hat{s}_M^*(\cdot)$. The proposal density in (9) can be approximated by replacing $s_M(\cdot)$ with $\hat{s}_M^*(\cdot)$. We repeat the procedure iteratively, with each iteration generating a new batch of samples, as suggested in (Li et al. 2021). The iterative IS-VS procedure is depicted in Figure 1 in the online supplementary materials.

4.4. Pre-screening procedure

When the input dimension is large, implementing Algorithm 1 could be computationally demanding. Thus, we propose a pre-screening procedure to reduce the number of candidate subsets of variables. The first pre-screening tool is the CE in (10). Note that candidates with high CE are likely to be ineffective, so we exclude them before the CV evaluation.

On the other hand, candidates with small CE could suffer from weight degeneracy, as discussed in Section 4.2. To avoid weight degeneracy, we employ the Pareto $\hat{k}$ diagnostic score (Vehtari et al.

Algorithm 1 Variable selection procedure in IS-VS.

- 1: **Input:** Pilot samples or data from previous iterations, I candidates of major variables $\mathbf{x}_M^{(i)}$ for $i = 1, \dots, I$.

- 2: Divide the available dataset into training set $\mathcal{D}_{n_{cv}}^{tr}$ and validation set $\mathcal{D}_{n_{cv}}^{vld}$ for $n_{cv} = 1, 2, \dots, N_{cv}$.

Phase 1:

- 3: **for** $i = 1, \dots, I$ **do**

- 4: **for** $n_{cv} = 1, 2, \dots, N_{cv}$ **do**

- 5: Set $\mathbf{x}_M = \mathbf{x}_M^{(i)}$.

- 6: Obtain $\hat{s}_M^{tr}(\mathbf{x}_M)$ using $\mathcal{D}_{n_{cv}}^{tr}$, e.g, applying (29) for nonparametric IS.

- 7: Construct $q_{IS-VS}^*(\mathbf{X})$ in (9) after replacing $s_M(\mathbf{x}_M)$ with $\hat{s}_M^{tr}(\mathbf{x}_M)$ in (9).

- 8: Sample $\mathbf{X}_j = (\mathbf{X}_{Mj}, \mathbf{X}_{-Mj})$ from $q_{IS-VS}^*(\mathbf{X})$ for $j = 1, 2, \dots, n_{tr}$.

- 9: Calculate $\tilde{P}_{tr}$ in (15).

- 10: Set $\hat{P}_{n_{cv}} = \tilde{P}_{tr}$.

- 11: **end for**

- 12: Set $\bar{P} = \frac{1}{N_{cv}} \sum_{n_{cv}} \hat{P}_{n_{cv}}$.

- 13: Obtain $SE(\tilde{P}_{tr})$ by $SE(\tilde{P}_{tr}) = \frac{1}{N_{cv}-1} \sum_{n_{cv}} (\hat{P}_{n_{cv}} - \bar{P})^2$.

- 14: **end for**

Phase 2:

- 15: **for** $i = 1, 2, \dots, I$ **do**

- 16: **if** $SE(\tilde{P}_{tr})$ satisfies (18)-(20) **then**

- 17: **for** $n_{cv} = 1, 2, \dots, N_{cv}$ **do**

- 18: Obtain $\hat{s}_M^{vld}(\mathbf{x}_M)$ using $\mathcal{D}_{n_{cv}}^{vld}$, e.g, applying (29) for nonparametric IS.

- 19: Construct $q_{IS-VS}^*(\mathbf{X})$ in (9) after replacing $s_M(\mathbf{x}_M)$ with $\hat{s}_M^{vld}(\mathbf{x}_M)$ in (9).

- 20: Sample $\mathbf{X}_j = (\mathbf{X}_{Mj}, \mathbf{X}_{-Mj})$ from $q_{IS-VS}^*(\mathbf{X})$ for $j = 1, 2, \dots, n_{vld}$.

- 21: Calculate $\tilde{P}_{vld}$ in (23).

- 22: Set $\hat{P}_{n_{cv}} = \tilde{P}_{vld}$.

- 23: **end for**

- 24: Set $\bar{P} = \frac{1}{N_{cv}} \sum_{n_{cv}} \hat{P}_{n_{cv}}$.

- 25: Obtain $SE(\tilde{P}_{vld})$ by $SE(\tilde{P}_{vld}) = \frac{1}{N_{cv}-1} \sum_{n_{cv}} (\hat{P}_{n_{cv}} - \bar{P})^2$.

- 26: **end if**

- 27: **end for**

- 28: Choose the best set $\mathbf{X}_M^*$ and its proposal density q_M^* with the smallest $SE(\tilde{P}_{vld})$.

- 29: **Output:** $\mathbf{X}_M^*$ and q_M^* .

2024), as our second pre-screening tool. To obtain the Pareto $\hat{k}$ score, we take the largest 20% of IS weights δ_j in (11) from the samples drawn from its proposal density and fit them with a Generalized Pareto Distribution. The distribution has $1/k$ finite moments, where k is the shape parameter. The estimated parameter $\hat{k}$, termed the Pareto $\hat{k}$ score, indicates the stability of the proposal density. It is suggested that the estimation is likely to be reliable and unbiased with finite variance when $\hat{k} < 0.7$. When $\hat{k} \geq 0.7$, the IS weights could be heavy-tailed, potentially causing degeneracy.

With these two tools, we detail the pre-screening procedure. Suppose that we consider N_{scn} pre-screened candidates for CV. For low to moderate dimensional cases, we compute the CE of all candidates and rank them by CE from the smallest to the largest. We then select the first N_{scn} candidates whose generated samples have a Pareto $\hat{k}$ score less than 0.7. For high-dimensional cases, we adopt a greedy strategy that is similar to the forward selection in regression. Starting from an empty set, we iteratively add the variable (or sub-estimator in the nonparametric construction discussed in Section 5) that yields the smallest CE, as long as its Pareto $\hat{k}$ score is less than 0.7. We continue this process until N_{scn} candidates are identified. Algorithm 2 summarizes the pre-screening procedure.

The proposed pre-screening procedure integrates the Pareto $\hat{k}$ diagnostic as a filtering criterion throughout the selection process, rather than relying solely on CE improvement. In our framework, CE serves as the guiding metric for selection, while the Pareto $\hat{k}$ score acts as a diagnostic filter to eliminate unstable candidates. This dual-criterion approach enables the pre-screening procedure to identify influential variables while mitigating the risk of weight degeneracy. After completing the pre-screening procedure, we apply the IS-VS algorithm in Algorithm 1 with the pre-screened candidates to determine $\mathbf{X}_M^*$.

4.5. Importance quantification and interpretation

This section quantifies the importance of each selected variable in $\mathbf{X}_M^*$ and interprets the results. Specifically, we employ a tool used for sensitivity analysis that explains the importance of each variable (Song et al. 2016). Among various approaches, variance-based analysis decomposes the variance of the output into components attributed to each input, measuring its contribution to the output variance (Sobol 2001, Jeong and Byon 2026). The Sobol index is one of the most widely used methods, however, it typically assumes independence among variables. Shapley effects, another popularly employed method in explainable AI, can calculate the importance of each variable even when variables are correlated (Song et al. 2016, Cohen et al. 2023). As input variables are correlated in our wind turbine case study in Section 7, we employ Shapley effects in our analysis.

Algorithm 2 Pre-screening procedure

```

1: Input:  $N_{scn}$ .
2: Case A: Moderate-dimensional input
3: Set  $n_{scn} = 0$ .
4: for each  $\mathbf{X}_M$  do
5:   Compute CE of the conditional POE estimator  $\hat{s}_M(\cdot)$ .
6: end for
7: Sort all subsets by CE (smallest first).
8: while  $n_{scn} < N_{scn}$  do
9:   for each sorted  $\mathbf{X}_M$  do
10:    Generate samples from its proposal density in (9).
11:    Compute the Pareto  $\hat{k}$  score of the generated samples.
12:    if  $\hat{k} < 0.7$  then
13:      Keep  $\mathbf{X}_M$  and set  $n_{scn} = n_{scn} + 1$ .
14:    end if
15:   end for
16: end while

17: Case B: High-dimensional input
18: Set  $n_{scn} = 0$ ,  $\mathbf{X}_M = \emptyset$ .
19: while  $n_{scn} < N_{scn}$  do
20:   Find the input variable  $X \notin \mathbf{X}_M$  that gives the best CE improvement if added.
21:   Set  $\mathbf{X}'_M = (\mathbf{X}_M, X)$  and compute its CE.
22:   Generate samples from the proposal density in (9) with  $\mathbf{X}'_M$ .
23:   Compute Pareto  $\hat{k}$  score of the generated samples.
24:   if  $\hat{k} < 0.7$  then
25:     Keep  $\mathbf{X}'_M$ .
26:     Set  $\mathbf{X}_M = \mathbf{X}'_M$  and  $n_{scn} = n_{scn} + 1$ .
27:   end if
28: end while

29: Output: Pre-screened subsets.

```

Specifically, we compute the marginal contribution of each variable X as follows.

$$\psi_X = \sum_{\mathcal{U} \subset \mathbf{X}_M^* \setminus \{X\}} \frac{|\mathcal{U}|!(d_M - |\mathcal{U}| - 1)!}{d_M!} (v(\mathcal{U} \cup X) - v(\mathcal{U})), \quad (24)$$

where $\mathcal{U}$ is any subset of $\mathbf{X}_M^*$ that does not contain X , and $v(\cdot)$ represents the value function. The term $\frac{|\mathcal{U}|!(d_M - |\mathcal{U}| - 1)!}{d_M!}$ is employed to provide an equal weight to any subset $\mathcal{U}$ with the same size and ensures that the weights sum up to 1. The value function $v(\cdot)$ explains the variability of the simulation output, and we set $v(\cdot)$ as

$$v(\mathcal{U}) = \text{Var} \left[E \left[\mathbb{I}(Y > l) | \mathbf{X}_{M,\mathcal{U}} \right] \right] = \text{Var} \left[P(Y > l) | \mathbf{X}_{M,\mathcal{U}} \right] = \text{Var} \left[s_{M,\mathcal{U}}(\mathbf{X}_{M,\mathcal{U}}) \right], \quad (25)$$

where $\mathbf{X}_{M,\mathcal{U}}$ is the variables in $\mathcal{U}$ and $s_{M,\mathcal{U}}(\cdot)$ is the conditional POE estimator with $\mathbf{X}_{M,\mathcal{U}}$. $v(\mathcal{U})$ implies how much $s_{M,\mathcal{U}}(\cdot)$ changes over the input space. The more influential $\mathbf{X}_{M,\mathcal{U}}$ is, the larger $v(\mathcal{U})$ we would expect. Note that this variance is different from the variance of the POE estimator, which we aim to minimize. In practice, $s_{M,\mathcal{U}}(\cdot)$ is unknown, so we use its estimator $\hat{s}_{M,\mathcal{U}}(\cdot)$ in our implementation. Additionally, we use the sample variance to estimate $v(\mathcal{U})$.

The difference between the two value functions in (24) represents the influence of the input $\mathbf{X}$ on the variability of the conditional POE across the input space. Influential variables tend to have large Shapley effects, as they significantly impact the conditional POE. In contrast, non-influential variables could have small Shapley effects. Small Shapley effects imply that adding $\mathbf{X}$ provides only a marginal contribution to explaining the variability in the conditional POE.

5. Nonparametric Construction

In implementing IS-VS, we need to estimate the conditional POE estimator $s_M(\cdot)$ based on $\mathbf{X}_M$. Several approaches can be considered, and the proposed variable selection framework can be integrated with various regression or classification methods. While developing a conditional POE estimator is not the primary focus of this paper, we adopt the non-parametric approach known as WAMK-SIS, which has demonstrated superior performance (Li et al. 2021). This section presents the proposed IS-VS with the WAMK-SIS method.

5.1. Stochastic importance sampling with weighted additive multiplicative kernel

We first review the WAMK-SIS procedure that uses the Nadaraya–Watson (N-W) kernel regression for estimating the conditional POE $s(\mathbf{x})$. Several kernel functions could be used, each with its own advantages and disadvantages. Seeking a balanced evaluation between the two extremes of fully additive and multiplicative kernels, Li et al. (2021) suggested a weighted additive bivariate kernel.

The use of bivariate kernels allows for the characterization of two-way interaction effects, under the assumption that higher order interactions are negligible. Specifically, WAMK-SIS employs the following estimator for $s(\mathbf{x})$.

$$\hat{s}(\mathbf{x})_{full} = \sum_{1 \leq p < q \leq d} w_{pq} B(x_p, x_q), \quad (26)$$

where w_{pq} denotes the weight of $B(x_p, x_q)$ such that $\sum_{1 \leq p < q \leq d} w_{pq} = 1$, and $B(x_p, x_q)$ is the N-W bivariate kernel estimator, defined by

$$B(x_p, x_q) = \frac{\sum_{j=1}^n K_2\left(\frac{x_p - X_{pj}}{h_p}, \frac{x_q - X_{qj}}{h_q}\right) Z_j}{\sum_{j=1}^n K_2\left(\frac{x_p - X_{pj}}{h_p}, \frac{x_q - X_{qj}}{h_q}\right)}, \quad (27)$$

with $Z_j = \mathbb{I}(Y_j > l | \mathbf{X} = \mathbf{X}_j)$, $\mathbf{X}_j = [X_{1j}, X_{2j}, \dots, X_{dj}]^T$, and X_{pj} and X_{qj} denoting the j -th samples of two variables X_p and X_q , respectively. Further, we have $\mathbf{x} = [x_1, x_2, \dots, x_d]^T$ and $K_2(\cdot, \cdot)$ is a 2-dimensional bivariate kernel, a product of univariate kernels of two variables, and h_p and h_q are the bandwidths of X_p and X_q , respectively, which can be obtained by minimizing the asymptotic mean integrated squared error. $B(x_p, x_q)$ is also called a subestimator.

5.2. Variable selection

While WAMK-SIS demonstrates the superiority over other IS methods (Li et al. 2021), it still has limitations in handling multivariate input as the input dimension d increases. Firstly, with larger d , the number of subestimators $B(\cdot, \cdot)$ in (26) increases significantly, specifically in the order of $\binom{d}{2}$. This increase results in unwarranted complexity in constructing the proposal distribution. Secondly, the full enumeration of bivariate subestimators could lead to redundant kernels. For instance, suppose X_2 and X_3 are of minimal significance and their interactions with X_1 are negligible; under these circumstances, $B(x_1, x_2)$ and $B(x_1, x_3)$ might closely resemble each other, that is, $B(x_1, x_2) \approx B(x_1, x_3)$. This indicates that $\hat{s}(\mathbf{x})_{full}$ in (26) could contain redundant terms, similar to the multicollinearity issue in the regression.

Our goal is to choose informative subestimators out of $\binom{d}{2}$ estimators. Therefore, the variable selection in the proposed IS-VS is translated into the selection of $B(\cdot, \cdot)$'s. Let Ω_M denote the set of variable indices of major subestimators. For instance, for the conditional POE estimator with $B(x_1, x_2)$ and $B(x_1, x_3)$, we have $\Omega_M = \{(1, 2), (1, 3)\}$. With the $\binom{d}{2}$ subestimators, there are $2^{\binom{d}{2}} - 1$ possible options to construct Ω_M . Our approach is to choose the best set of subestimators

by using the complexity measure in (12). For the penalty term, we replace $\mathcal{P}(d_M)$ with the number of subestimators in Ω_M , i.e., its cardinality $|\Omega_M|$. Thus the complexity measure becomes

$$C_\alpha(\Omega_M) = SE(\hat{P}_M) + \alpha|\Omega_M|. \quad (28)$$

To obtain $SE(\hat{P}_M)$, we define the conditional POE estimator $\hat{s}_M(\mathbf{x}_M)$ as follows.

$$\hat{s}_M(\mathbf{x}_M) = \sum_{(p,q) \in \Omega_M} w_{pq} B(x_p, x_q), \quad (29)$$

where w_{pq} quantifies the importance of $B(x_p, x_q)$ as

$$w_{pq} = \frac{1/e(p, q)}{\sum_{1 \leq p' < q' \leq d} 1/e(p', q')}. \quad (30)$$

Here, $e(p, q)$ implies the error for $B(x_p, x_q)$ as follows

$$e(p, q) = - \sum_{j=1}^n \{Z_j \ln(B(X_{pj}, X_{qj})) + (1 - Z_j) \ln(1 - B(X_{pj}, X_{qj}))\}. \quad (31)$$

Note that $e(p, q)$ diminishes as $B(x_p, x_q)$ approaches Z_j more closely, similar to the CE function in (10). In this sense, a smaller $e(p, q)$ indicates good estimation of $B(x_p, x_q)$ (Li et al. 2021).

With large d , evaluating the performance of each candidate through the proposed CV procedure could be computationally burdensome. We pre-screen the candidates of subestimators using Algorithm 2 by replacing $\mathbf{X}_M$ with Ω_M . In practice, we adopt the moderate-dimensional case of Algorithm 2 when $d \leq 5$, and the high-dimensional case when $d > 5$.

Finally, we find the best Ω_M^* by utilizing the variable selection procedure discussed in Section 4.3. For quantifying the importance of each selected subestimator and interpreting the results, we employ the sensitivity index in (24) and consider the marginal contribution of each subestimator b to a value function as follows.

$$\psi_b = \sum_{\mathcal{U} \subset \Omega_M \setminus \{b\}} \frac{|\mathcal{U}|!(|\Omega_M| - |\mathcal{U}| - 1)!}{|\Omega_M|!} (v(\mathcal{U} \cup b) - v(\mathcal{U})). \quad (32)$$

6. Numerical Examples

This section conducts numerical experiments to evaluate the effectiveness of the IS-VS approach.

6.1. Problem setting

We consider four examples with $Y|X = \mathbf{x} \sim N(\mu_m(\mathbf{x}), 1)$, for $m = 1, 2, 3, 4$, where m denotes the example number. The conditional mean $\mu_m(\mathbf{x})$ of Y , given $\mathbf{X}$, in each example is as follows.

$$\begin{aligned}\mu_1(\mathbf{x}) &= 65 - 40e^{-0.2\sqrt{(x_1^2+x_2^2)/2}} - 20e^{-0.2|x_1|} - 5e^{-0.2\sqrt{(x_2^2+x_3^2+x_4^2)/3}} - \sum_{1 \leq i < j \leq 3} e^{\cos(2\pi x_i x_j)} \\ \mu_2(\mathbf{x}) &= 65 - 40e^{-0.2\sqrt{(x_1^2+x_2^2)/2}} - 20e^{-0.2|x_1|} - 5e^{-0.2\sqrt{(x_2^2+x_3^2)/2}} - 0.1e^{-0.2\sqrt{(x_4^2+x_5^2)/2}} \\ &\quad - \sum_{1 \leq i < j \leq 3} e^{\cos(2\pi x_i x_j)} \\ \mu_3(\mathbf{x}) &= 65 - 40e^{-0.2\sqrt{(x_1^2+x_2^2)/2}} - 20e^{-0.2|x_1|} - 5e^{-0.2\sqrt{(x_2^2+x_3^2)/2}} - 0.1(e^{-0.2\sqrt{(x_4^2+x_5^2)/2}} \\ &\quad + e^{-0.2\sqrt{(x_6^2+x_7^2)/2}} + e^{-0.2\sqrt{(x_8^2+x_9^2+x_{10}^2)/3}}) - \sum_{1 \leq i < j \leq 3} e^{\cos(2\pi x_i x_j)} \\ \mu_4(\mathbf{x}) &= 65 - 40e^{-0.2\sqrt{(x_1^2+x_2^2)/2}} - 20e^{-0.2|x_1|} - 5e^{-0.2\sqrt{(x_2^2+x_3^2)/2}} - 0.1(e^{-0.2\sqrt{(x_4^2+x_5^2)/2}} \\ &\quad + e^{-0.2\sqrt{(x_6^2+x_7^2)/2}} + e^{-0.2\sqrt{(x_8^2+x_9^2+x_{10}^2)/3}}) - \sum_{1 \leq i < j \leq 3} e^{\cos(2\pi x_i x_j)} - \sum_{i=11}^{50} 0.01e^{-0.2|x_i|}\end{aligned}$$

In Example 1, the 4-dimensional input variables are *i.i.d.* $N(0, \mathbb{I}_4)$, where $\mathbb{I}_4$ implies the 4×4 identity matrix (Li et al. 2021). Example 2 slightly modifies the structure of Example 1 with the 5-dimensional dependent input; while the distribution of X_1 , X_4 , and X_5 are independent and follow $N(0, \mathbb{I}_3)$, the distributions of X_2 and X_3 depend on X_1 such that $X_2, X_3|X_1 = x_1 \sim N(x_1, 1)$. The input variables in Example 3 are 10-dimensional and *i.i.d.* $N(0, \mathbb{I}_{10})$, taken from (Park et al. 2025). Finally, Example 4 represents a higher-dimensional case with 50 input variables where the inputs are assumed to be *i.i.d.* $N(0, \mathbb{I}_{50})$. We consider the failure probability $P(Y > l_m)$ with $l_1 = 18.99$, $l_2 = 24.2$, $l_3 = 18.74$, and $l_4 = 18.42$, which correspond to the failure probability of 0.01 in all examples.

We apply our IS-VS procedure using the nonparametric IS construction discussed in Section 5. We take an iterative procedure with a total budget of 1200 samples. Specifically, we first use the Latin hypercube sampling to generate 200 pilot samples. Then we iterate the procedure five times, with each iteration generating 200 samples, similar to the setting in Li et al. (2021). The POE of each experiment is the average POE over these five iterations. We conduct a total of 25 experiments for each example. We set $N_{cv} = 10$ in Algorithm 1 and $N_{scn} = 8$ in Algorithm 2.

We compare IS-VS with a few variants of IS-VS. One such variant selects the variables with the smallest CE value, referring to as IS-CE. Another approach, termed IS-Pareto, chooses the variables with the smallest CE among the candidates with Pareto $\hat{k}$ scores less than 0.7. By employing the Pareto $\hat{k}$ score, IS-Pareto aims to maintain stability without excessively narrowing its focus, potentially enhancing robustness compared to IS-CE. In these methods, we employ the same iterative procedure as in IS-VS.

Additionally, we consider state-of-the-art variance reduction methods. The first alternative is WAMK-SIS (Li et al. 2021), which employs all possible subestimators without variable selection. Next, the Entropy-based Contour Locator (ECL) method employs a Gaussian Process (GP) as a low fidelity model, selects a subset of samples corresponding to failures in the low fidelity model, and then evaluates the samples in the high fidelity model. By guiding the sampling process with the GP surrogate, ECL aims to generate more outputs corresponding to failure events, thereby enhancing the efficiency of the sampling process (Cole et al. 2023). Further, we compare our method with the stratified sampling with selected variables (Park et al. 2025). This method, called OptiTreeStrat, constructs strata using a binary tree and allocates sample sizes across strata according to their estimated importance, a strategy analogous to reweighting sampling efforts in IS.

We also compare our approach with three existing dimension reduction approaches that can provide the conditional POE estimate $\hat{s}_M(\mathbf{x}_M)$ based on a selected subset $\mathbf{X}_M$ of input variables. (1) Lasso-regularized logistic regression (Lasso-logit) incorporates an ℓ_1 penalty into logistic regression to induce sparsity in the coefficient vector (Tibshirani 1996). By shrinking small coefficients to zero, it performs variable selection. (2) Sparse additive model (SpAM) is a nonparametric regression approach that formulates the response as a sparse sum of smooth, low-dimensional nonlinear component functions (Ravikumar et al. 2009). The sparsity constraint shrinks many component functions to zero, thereby enabling variable selection while allowing flexible nonlinear effects for the selected predictors; (3) Random forest with recursive feature elimination (RF-RFE) is a wrapper-based method that repeatedly eliminates variables deemed least important, as determined by random forest importance scores. This approach is effective for uncovering key predictors in high-dimensional, nonlinear data (Darst et al. 2018). In each of these three approaches, we first identify the major variables $\mathbf{X}_M$, obtain the conditional POE estimate $\hat{s}_M(\mathbf{x}_M)$, and then substitute $s_M(\mathbf{x}_M)$ into the proposal density (9) to draw samples and conduct simulations.

In each experiment, we use the same pilot samples and seeds for IS-VS, WAMK-SIS, IS-CE, IS-Pareto, and the three dimension reduction approaches. For ECL and OptiTreeStrat, we use the

different pilot sample sizes in accordance with their respective suggested rules. Across all methods, we use the same total computational budget of 1,200 to ensure a fair comparison. We evaluate the mean and SE of the POE estimates obtained from 25 experiments for each approach.

6.2. Comparison results

Table 1 reports the implementation results from 25 experiments. Below we summarize our observations.

- IS-VS achieves the smallest SE compared to all alternative methods in all cases, with its average POE estimates being close to the true POE of 0.01. It is noteworthy that, in the high-dimensional example (Example 4), the proposed IS-VS method achieves the smallest SE, whereas the existing three dimension reduction approaches, Lasso-logit, SpAM, and RF-RFE, perform substantially worse. These results provide compelling evidence of the superiority and scalability of our approach in tackling high-dimensional problems.
- WAMK-SIS, while outperforming some alternatives, yields larger SEs compared to IS-VS, because its full use of all bivariate estimators prevents effective concentration.
- The poor performance of IS-CE demonstrates that concentrating simulation efforts on a small number of selected variables can be counterproductive when the concentration is overly restrictive. Due to its large SEs, the resulting average POE estimates significantly deviate from the true POE of 0.01. In many cases, IS-CE generates large Pareto $\hat{k}$ scores exceeding 0.7.
- IS-Pareto attempts to mitigate the degeneracy issue by choosing the estimator with a small or moderate Pareto $\hat{k}$ score. With these efforts, its performance improves over IS-CE, however, its performance still remains much worse compared to IS-VS. This highlights the advantage of the newly devised complexity measure and CV procedure in IS-VS, which selects the variables with the minimum out-of-sample variance.
- ECL achieves smaller SEs in Examples 2 and 3 compared to WAMK-SIS; however, its SEs are still larger than those of IS-VS. Another major shortcoming of ECL is its extensive computation time. ECL takes nearly ten times longer than IS-VS for each example. For instance, without parallel computing, IS-VS requires around 5 hours to complete 25 experiments for Example 1, whereas ECL takes nearly 2 days for the same number of experiments.
- OptiTreeStrat achieves the second-best performance among all methods after IS-VS, highlighting the advantage of its variable selection capability. However, its binary tree construction makes locally optimal choices at each partition while recursively growing the tree. Consequently, its

performance is highly dependent on the initial partitions, and poor initial splits can lead to ineffective strata design.

- The three dimension reduction methods exhibit larger SEs than IS-VS in all four examples. In particular, SpAM yields POE estimates below 0.01 in some cases, because it often generates very few samples satisfying $Y > I$, leading to an underestimation of POE and a correspondingly large variance. The mean estimates obtained from RF-RFE also deviate considerably from 0.01, likely due to weight degeneracy, as this method frequently produces large Pareto $\hat{k}$ scores.

Table 1 Comparison results in numerical examples from 25 experiments (note: The best results with the smallest SE are highlighted in bold font; The results with ECL could not be obtained for Example 4 within a reasonable time and are therefore omitted.)

	Example 1		Example 2		Example 3		Example 4	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE
IS-VS	0.0098	0.0013	0.0101	0.0011	0.0096	0.0013	0.0101	0.0012
IS-CE	0.0231	0.0057	0.0240	0.0081	0.0232	0.0061	0.0107	0.0015
IS-Pareto	0.0128	0.0059	0.0206	0.0079	0.0152	0.0045	0.0107	0.0015
WAMK-SIS	0.0098	0.0020	0.0096	0.0022	0.0092	0.0027	0.0102	0.0024
ECL	0.0100	0.0023	0.0097	0.0014	0.0098	0.0016	N/A	N/A
OptiTreeStrat	0.0103	0.0015	0.0096	0.0015	0.0096	0.0016	0.0101	0.0026
Lasso-logit	0.0102	0.0020	0.0094	0.0027	0.0094	0.0025	0.0097	0.0030
SpAM	0.0089	0.0058	0.0074	0.0053	0.0104	0.0058	0.0086	0.0044
RF-RFE	0.0212	0.0025	0.0393	0.0055	0.0205	0.0028	0.0210	0.0029

Figure 1 depicts violin plots comparing the distribution of IS weights in (11) in one of the experiments. The width of each curve corresponds with the frequency of weights. Most IS weights of IS-VS are small, helping to avoid weight degeneracy, and they have similar contributions toward the POE estimation. On the contrary, some weights in IS-CE are very large and dominate smaller ones when computing the POE in (11), inflating variance and causing substantial deviations of the estimate from the true value.

Next, we conduct experiments using different conditional POE estimators, rather than the non-parametric WAMK construction discussed in Section 5. Specifically, we consider two alternative estimation approaches, RF and GP, which represent widely used nonparametric and probabilistic methods for conditional probability estimation. As shown in Table 2, IS-VS with WAMK (denoted

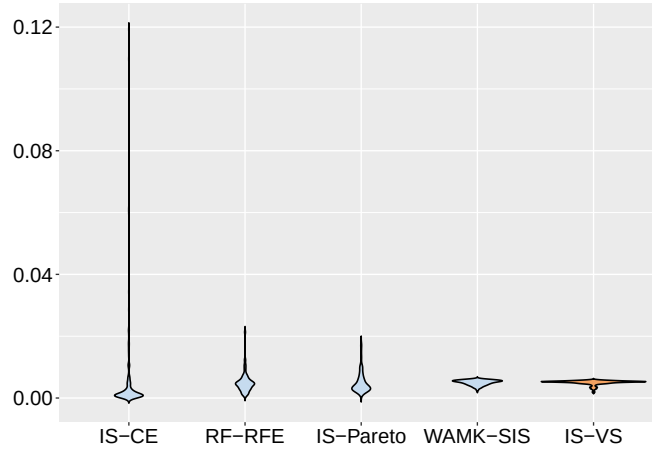


Figure 1 Distribution of IS weights in one of the experiments for Example 1

as IS-VS (WAMK)), consistently achieves smaller SEs compared with IS-VS with RF and GP (denoted as IS-VS (RF) and IS-VS (GP), respectively) across all examples.

Table 2 Comparison results with different conditional POE estimators in IS-VS from 25 experiments (note: The best results with the smallest SE are highlighted in bold font; The results with RF could not be obtained within a reasonable time in Example 2 due to the extremely low acceptance rate in acceptance–rejection sampling and are therefore omitted.)

	Example 1		Example 2		Example 3		Example 4	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE
IS-VS (WAMK)	0.0098	0.0013	0.0101	0.0011	0.0096	0.0013	0.0101	0.0012
IS-VS (RF)	0.0174	0.0023	N/A	N/A	0.0162	0.0028	0.0157	0.0029
IS-VS (GP)	0.0109	0.0032	0.0105	0.0017	0.0098	0.0015	0.0104	0.0020

More details on the dimension reduction results achieved by IS-VS are provided in the online supplementary materials. Furthermore, additional experiments using the IS-VS approach without pre-screening were conducted, and the corresponding results are also reported in the online supplementary materials.

6.3. Importance quantification

To further examine whether IS-VS can successfully identify the important variables, we report in Table 3 the selection frequency (percentage) of each subestimator based on 25 experiments for Example 1, where X_1 is the most influential, followed by X_2 . Naturally, the most critical subestimator $B(x_1, x_2)$ is always included in all experiments. In fact, IS-CE often chooses this subestimator only;

however, using it alone would lead to high variance, as shown in Table 1. On the contrary, IS-VS includes other subestimators, such as $B(x_3, x_4)$ and $B(x_2, x_4)$, in most cases. These additional kernel estimators include less important variables to facilitate sufficient exploration. We observe similar results for other examples but omit them to save space.

Table 3 Percent of frequently selected bivariate subestimators in IS-VS for Example 1.

Bivariate Kernel	Percent
$B(x_1, x_2)$	100%
$B(x_3, x_4)$	30.4%
$B(x_2, x_4)$	26.4%
$B(x_2, x_3)$	25.6%
$B(x_1, x_4)$	10.4%
$B(x_1, x_3)$	6.4%

Lastly, we quantify the importance of each subestimator and interpret the results using the Shapley effect analysis. The first column of Table 4 lists some of the most frequently selected subestimators in IS-VS, and the last column reports their average Shapley effects for Example 1. For instance, when $\Omega_M = \{(1, 2), (3, 4)\}$ is chosen, $B(x_1, x_2)$ has significantly larger Shapley effects, demonstrating its strong influence on the simulation output and its role for concentrating simulation efforts. On the other hand, $B(x_3, x_4)$ is included to facilitate adequate exploration and to prevent possible weight degeneracy. Likewise, $B(x_1, x_2)$ often receives the highest Shapley effect in other instances, while additional kernels are included to promote exploration. We observe similar results for other examples.

7. Wind Turbine Case Study

One significant application of digital computer models lies in the evaluation of wind turbine reliability. As wind turbines increase in size, the importance of reliability testing has grown substantially (Ding 2019). The National Renewable Energy Laboratory (NREL), under the U.S. Department of Energy (DOE), has developed aeroelastic computer models (Jonkman 2009, Jonkman and Buhl Jr 2005). Specifically, we use the NREL's simulators, including TurbSim (version 1.50) (Jonkman 2009) and FAST (version 7.01.00a-bjj) (Jonkman and Buhl Jr 2005).

We consider the 10-minute maximum tip deflection as the response variable Y , which is one of the essential blade load responses related to the wind turbine stability (Choe et al. 2016,

Table 4 Examples of Shapley effects for Example 1 (note: The negative Shapley effect values may arise from sampling noise due to the limited sample size, as well as approximation errors in $\hat{s}_M(\cdot)$)

Ω_M	Subestimator	Average Shapley effect
$B(x_1, x_2), B(x_3, x_4)$	$B(x_1, x_2)$	0.9895
	$B(x_3, x_4)$	0.0105
$B(x_1, x_2), B(x_2, x_3)$	$B(x_1, x_2)$	0.9007
	$B(x_2, x_3)$	0.0993
$B(x_1, x_2), B(x_2, x_4)$	$B(x_1, x_2)$	0.9808
	$B(x_2, x_4)$	0.0192
$B(x_1, x_2), B(x_1, x_4), B(x_2, x_4)$	$B(x_1, x_2)$	0.7199
	$B(x_1, x_4)$	0.3845
	$B(x_2, x_4)$	-0.1044

Mishnaevsky Jr 2022). We set the resistance level at $l = 2.06$. We consider the five input variables as in Li et al. (2021), including the wind speed V , turbulence intensity TI , wind shear S , wind vertical angle VA , and surface roughness length, SR . Therefore, we have $\mathbf{X} = (V, TI, S, VA, SR)$. For their original input distribution f , we use the same setting as outlined in Li et al. (2021). We conduct 25 experiments. Each experiment draws 600 pilot samples and includes 10 iterations where each iteration generates 200 samples from the NREL simulators.

7.1. Implementation results

We compare our approach IS-VS with the WAMK-SIS and OptiTreeStrat. We omit the implementation of ECL, because of its excessive computation time. We also exclude the performance of other approaches due to their poor results in the numerical examples in Section 6. Table 5 summarizes the results. The average POE estimates obtained from 25 experiments are comparable across all three methods. However, IS-VS achieves the smallest SE, reducing it by more than 37% and 46% compared to WAMK-SIS and OptiTreeStrat, respectively.

Table 5 Comparison results from 25 experiments

	Mean	SE
IS-VS	0.0495	0.0022
WAMK-SIS	0.0495	0.0035
OptiTreeStrat	0.0478	0.0041

Table 6 presents the percentage of bivariate kernel subestimators selected in IS-VS. In approximately half of the experiments, only two subestimators are used to construct the conditional POE. In other instances, IS-VS selects three to four subestimators, which is considerably fewer than the ten subestimators used in WAMK-SIS.

Table 6 Percentage of the number of bivariate kernel subestimators employed in IS-VS

Number of bivariate kernels	2	3	4
Percentage	51.2%	48.4%	0.4%

Table 7 lists the most frequently selected subestimators contained in Ω_M and their Shapley effects. In every experiment, the subestimator $B(V, TI)$ is included, because V and TI are the most important variables and their interaction is significant (Li et al. 2021). The importance of each subestimator is quantified by Shapley effects, as shown in the last column, where $B(V, TI)$ consistently exhibits the largest Shapley effect. While different subestimators are also included across various experiments due to the stochastic nature of stochastic simulations, their roles remain consistent. Their small Shapley effect values indicate their function as supplementary elements, helping to prevent excessive concentration on narrow regions and promoting adequate exploration across the input space.

Table 7 Frequently selected bivariate subestimators and their Shapley effects in wind turbine case study
(note: The negative Shapley effect values may arise from sampling noise due to the limited sample size, as well as approximation errors in $\hat{s}_M(\cdot)$)

Ω_M	Subestimator	Average Shapley effect
$B(V, TI), B(S, VA)$	$B(V, TI)$	1.3641
	$B(S, VA)$	-0.3641
$B(V, TI), B(TI, S)$	$B(V, TI)$	1.2169
	$B(TI, S)$	-0.2169
$B(V, TI), B(V, S), B(S, VA)$	$B(V, TI)$	1.1025
	$B(V, S)$	0.2880
	$B(S, VA)$	-0.3906
$B(V, TI), B(V, S), B(TI, S)$	$B(V, TI)$	1.0393
	$B(V, S)$	0.2896
	$B(TI, VA)$	-0.3289

Lastly, since $B(V, TI)$ is always included in Ω_M across all experiments, we would like to reiterate the risks of including only these dominant variables in Ω_M , i.e., $\Omega_M = \{(V, TI)\}$. In fact, IS-CE often leans towards this choice. Figure 2 compares the IS weights from IS-CE and IS-VS from one experiment. The weights of IS-VS are mostly consistent, whereas the range of weights in IS-CE is significantly wider, with some exceeding 0.3. These large weights can negatively influence the POE estimation, leading to weight degeneracy. The results demonstrate that coupling the seemingly less important subestimators with the most important $B(V, TI)$ is crucial to effectively bypass the degeneracy issue and to avert the potential problem of focusing excessively on narrow regions.

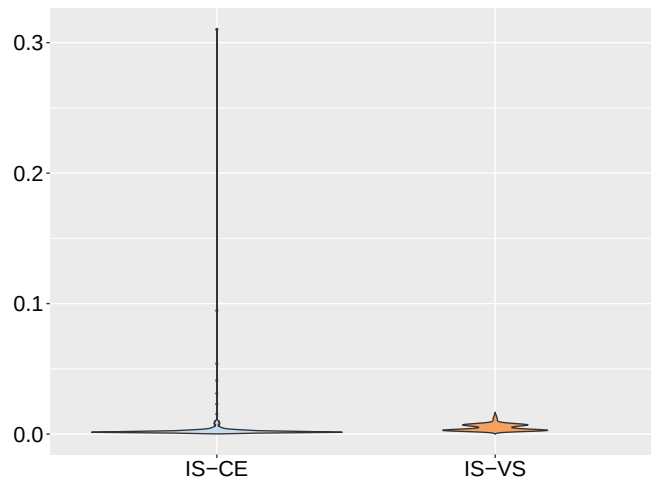


Figure 2 Distribution of IS weights from an experiment in wind turbine case study

8. Summary

This study devises a new IS method with multivariate inputs, focusing primarily on dimension reduction by selecting informative variables. We explore the risk of over-concentration and the trade-off between concentration and exploration. To achieve the balance, we introduce a new complexity measure and present a CV procedure, both specifically designed for variable selection in the IS context. Our CV procedure avoids additional computational burden of generating new data samples while effectively approximating variance in unseen datasets, guiding the selection of suitable variables. We interpret the results and quantify the importance of selected input variables through sensitivity analysis. Our analysis highlights the different roles of selected variables: some concentrate simulation efforts, while others facilitate exploration.

The practicality of employing the proposed method hinges on the duration of the simulation runs and the computational resources available. Our approach is well-suited for low- to mid-fidelity

simulators where each run takes minutes. For instance, in the wind turbine case study, simulating 10-minute operations of a land-based turbine takes about one minute. Conducting reliability analysis with high-fidelity simulators that have longer run times is a topic we plan to explore in future research. We also plan to investigate the variance reduction with multi-fidelity computer models.

9. Acknowledgements

The authors thank the Editor in Chief, senior editor, associate editor, and reviewers for their constructive and thoughtful comments on various aspects of this work.

10. Funding

This work was partly supported by the U.S. National Science Foundation (Grant No. CMMI-2226348).

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